

Course 1: Generalities about AI



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

2026-09-15

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Welcome to the first lesson of the Introduction to AI course!

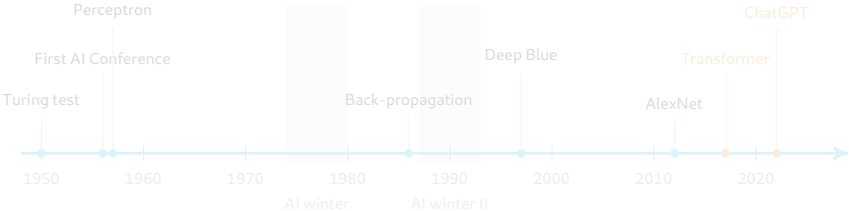
Artificial Intelligence?

A tentative definition of intelligence

- **Intelligence:** ability to **extract knowledge** from observations
- This knowledge is used to **solve tasks in different contexts and environments** (automation)

So AI: systems that solve tasks automatically. This definition is **very broad!**

AI is not a recent field!



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Intelligence: the ability to process information in order to inform future decisions.

AI is not new, and has gone several times through hype and disappointment. The first AI winter was in the 1970s, when the limitations of early symbolic AI became apparent. The second AI winter was in the late 1980s and early 1990s, when the limitations of expert systems and rule-based approaches became clear.

The recent trend of AI is rooted in deep learning, since around 15 years, and the very recent success of large language models (LLMs) and generative AI has brought AI into the mainstream.

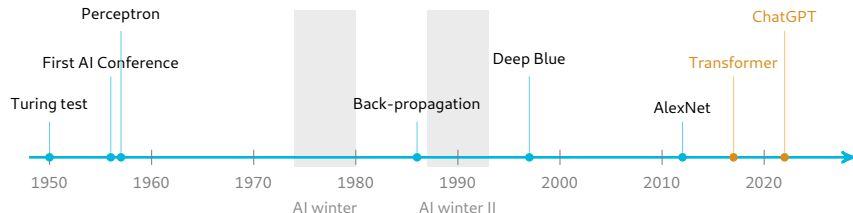
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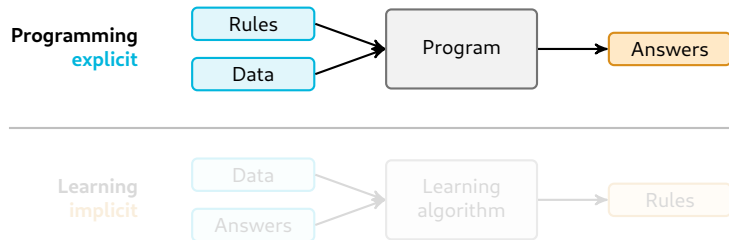


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Two ways of automation



A human writes the rules

- You know exactly what happens (exact, verifiable, debuggable)
- You must be able to **state** the rules

The machine infers the rules

- Works when nobody can state them
- Needs data, and the rules stay implicit (hard to audit)

AI is generally understood as the second way: learning from data.

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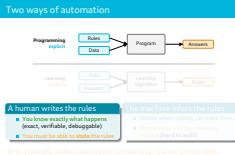
Two ways of automation

Terminology matters here, so let us be precise. The first way is **programming**, i.e. a human states the rules and the machine executes them. Sorting a list, computing a payroll: nobody trains a model for that, and nobody should.

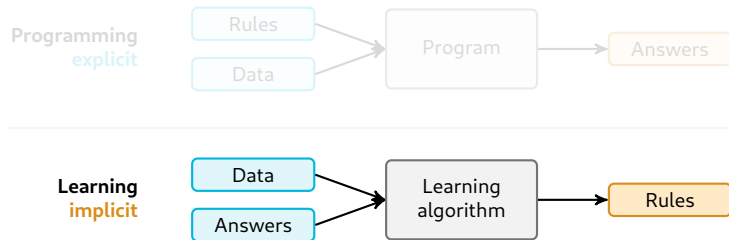
The second way is **learning**: instead of the rules, you provide examples of the answers, and the algorithm produces the rules (the model).

The inversion is the whole point: in programming, rules are an input and answers an output. In learning, answers are an input and rules the output. This is the explicit / implicit distinction, and it is also where the difficulties of this course come from: an implicit rule cannot be read, only tested.

(Diagram after F. Chollet, *Deep Learning with Python*, ch. 1.)



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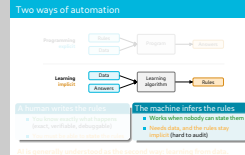
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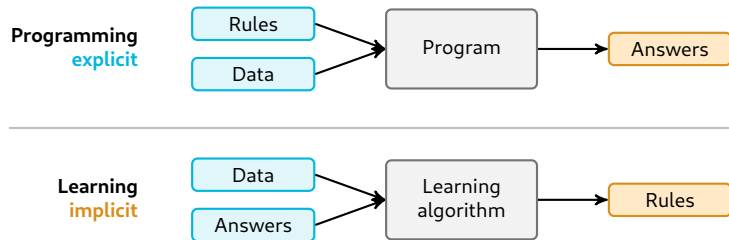
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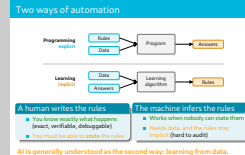
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"Learning", defined formally

A program **learns** from experience E with respect to a task T and a performance measure P , if its performance at T , measured by P , **improves with** E . (Tom Mitchell, 1997)

T : the **task**
what you predict

E : the **data**
what you learn from

P : the **metric**
how you are judged

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└ "Learning", defined formally

A definition that is worth memorising, because it is operational: it tells you what to write down before writing any code. Notice what it does *not* say: nothing about intelligence, nothing about the brain, nothing about neural networks.

Use it as a checklist on any AI claim you read, including your own project: what is the task, what data, and measured how? A system whose performance measure is unstated cannot be said to work.

Task, data and metric are precisely how you define your own project, and are the three required items of your validation presentation in session 2.

"Learning", defined formally

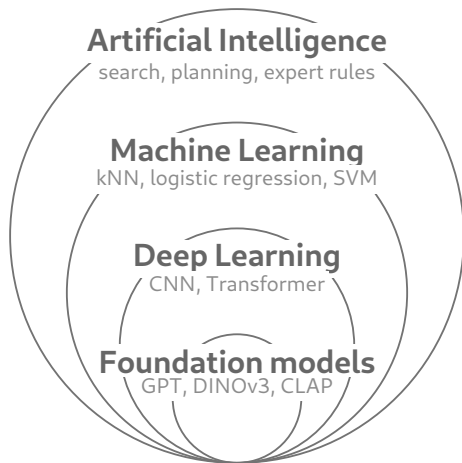
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AI, Machine Learning & Deep Learning

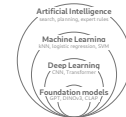


Adapted from *Deep Learning*, 2016 by A. Courville, I. Goodfellow and Y. Bengio

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AI, Machine Learning & Deep Learning



Adapted from Deep Learning, 2016 by A. Courville, I. Goodfellow and Y. Bengio

A useful way to understand the relationship between AI, Machine Learning, and Deep Learning.

Note that even if AI is broader than Machine Learning, the term AI is often used to refer to Machine Learning in practice (or even Deep Learning or Foundation models).

Traditional application domains of AI



Vision

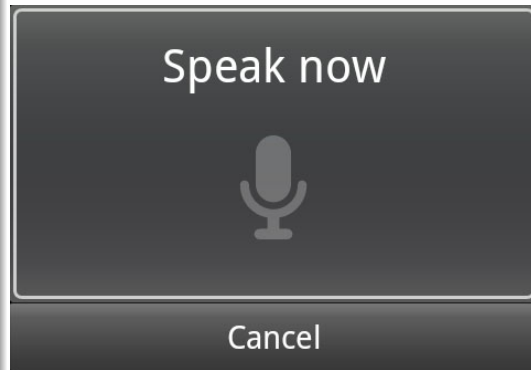
- Semantic Segmentation,
- Object/face recognition,
- Autonomous vehicles,
- Automatic diagnostic,
- Defects identification...



Vision is historically one of the first and most successful application domains of AI: recognizing objects or faces, detecting anomalies, and letting vehicles perceive their environment all build on the same underlying techniques.

Natural Language Processing (NLP)

- Automatic assistant (agents),
- Voice-to-text,
- Automatic translation,
- Automatic summarizing,
- Sentiment analysis,
- Text indexing...



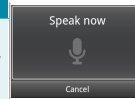
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Traditional application domains of AI

Natural Language Processing (NLP)

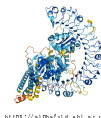
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Natural Language Processing covers everything related to understanding and generating text or speech: virtual assistants, translation, summarization, and sentiment analysis are all everyday examples.

Tons of other domains...

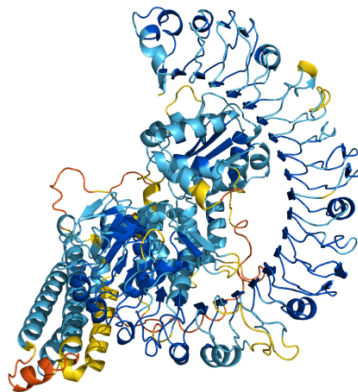
- Medical imaging,
- Games (chess, go, video games...),
- Audio and Music processing,
- Bioinformatics (AlphaFold),
- Recommender systems,
- Market analysis...



<https://alphafold.ebi.ac.uk/>

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Traditional application domains of AI

There are probably many more application domains than we can list here. As a striking example of a breakthrough, it is worth mentioning AlphaFold, which has revolutionized the field of bioinformatics. The AlphaFold network directly predicts the 3D coordinates of all heavy atoms of a given protein from its primary amino acid sequence. It is an AI system that combines many "ingredients" of modern Deep Learning approaches: attention mechanisms, self-distillation, and more.

Generative AI: a recent breakthrough

Text Synthesis with Large Language Models (LLMs)

By now, you should have all heard about LLMs: ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), LLaMA (Meta), Mistral 8x7B (MistralAI), ... And many others!

 Write an introduction to a master course on artificial intelligence for an engineering school



Welcome to the Master's course on Artificial Intelligence (AI) at our engineering school. In this program, you will gain a comprehensive understanding of the latest advances in the field of AI and machine learning. You will learn to design and implement intelligent systems using a variety of techniques and algorithms, and you will learn to critically evaluate the performance and limitations of these systems. The course will cover a wide range of topics including supervised and unsupervised learning, deep learning, natural language processing, computer vision, and robotics. You will also have the opportunity to work on real-world projects and to collaborate with other students and faculty members on cutting-edge research. By the end of the program, you will have the skills and knowledge needed to excel in a career in AI or to pursue further research in the field.

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└ Generative AI: a recent breakthrough

Does not require any more introduction: ChatGPT is the most famous example of a generative AI model, able to generate text in response to a prompt. It is based on a large language model (LLM) trained on a massive corpus of text data. We will cover the principles behind LLMs in session 5.

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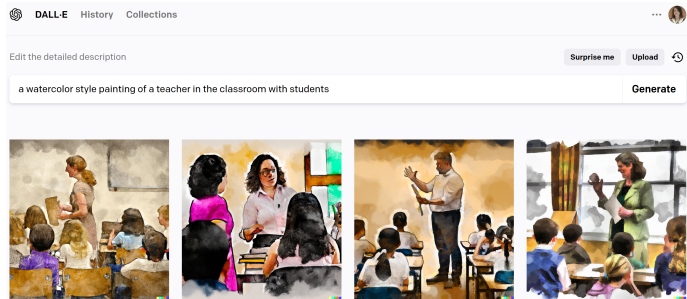


Generative AI: a recent breakthrough

Image Synthesis

- Stable Diffusion Models (<https://arxiv.org/pdf/2403.03206>)
- DALL-E (<https://labs.openai.com/>)
- Midjourney (<https://docs.midjourney.com/>)

Possible thanks to **multimodality**: manipulating simultaneously both text and image!

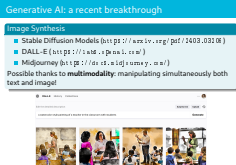


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└ Generative AI: a recent breakthrough

Multimodality allows for the simultaneous processing of multiple types of data, such as text and images, enabling more complex and nuanced AI applications.



Generative AI: a recent breakthrough

Many other modalities...

- Speech (Many actors, ex: ElevenLabs, Kyutai)
- Music (StableAudio, Udio, Suno, ...)
- Video (StableVideo, Gemini Omni, ...)
- ...
- **Multimodal** models are arising

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└ Generative AI: a recent breakthrough

Generative AI is not limited to text and images: the same diffusion- or transformer-based principles are now applied to speech, music, and video. Multimodal models, able to jointly handle several of these modalities at once, are an active and fast-moving research direction.

Generative AI: a recent breakthrough

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Some key open challenges (core AI research)

Interpretability



A trained model might be very accurate, but how does it take its decision ?

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Interpretability



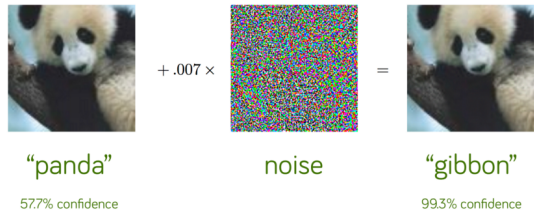
A trained model might be very accurate, but how does it take its decision ?

A trained model can reach very high accuracy while remaining a "black box": we often cannot explain precisely which cues it relies on to reach a decision. This matters a lot in high-stakes domains such as medicine, where clinicians need to trust and verify the model's reasoning.

The cow example here illustrates a case where the model has learned to recognize cows based on the presence of grass in the background, rather than the cow itself. This is a classic example of a spurious correlation, where the model has picked up on an irrelevant feature that happens to be correlated with the target class in the training data. Without interpretability tools, we would not know that the model is relying on this spurious feature, and it would fail to generalize to new images of cows in different contexts.

Some key open challenges (core AI research)

Learning what should be learned (robustness / adversarial attacks)



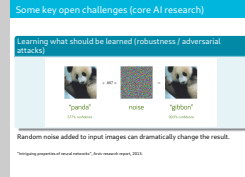
Random noise added to input images can dramatically change the result.

"Intriguing properties of neural networks", Arxiv research report, 2013.

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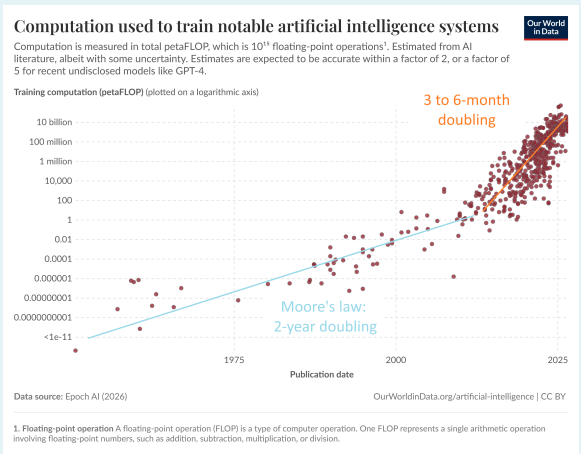
Some key open challenges (core AI research)



Neural networks can be surprisingly fragile: adding a small, carefully crafted amount of noise to an input image, imperceptible to a human, can make the model output a completely wrong prediction with high confidence. This raises serious concerns about robustness and security.

Some key open challenges (core AI research)

Computational and memory footprints

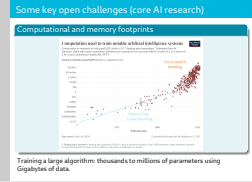


Training a large algorithm: thousands to millions of parameters using Gigabytes of data.

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└ Some key open challenges (core AI research)

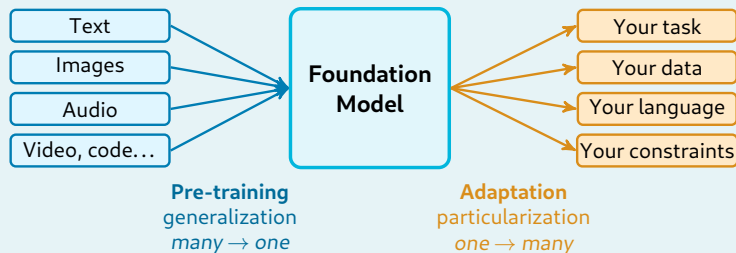


Training modern models is extremely resource-intensive: state-of-the-art networks have millions to billions of parameters, trained on gigabytes to terabytes of data, which raises practical questions about cost, energy consumption, and accessibility.

Some key open challenges (core AI research)

Yesterday's challenge: make **one** model *generalize* beyond its training set.

From generalization to particularization



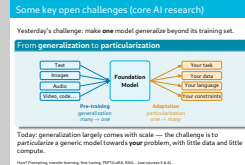
Today: generalization largely comes with scale — the challenge is to *particularize* a generic model towards **your** problem, with little data and little compute.

How? Prompting, transfer learning, fine-tuning, PEFT/LoRA, RAG... (see courses 5 & 6).

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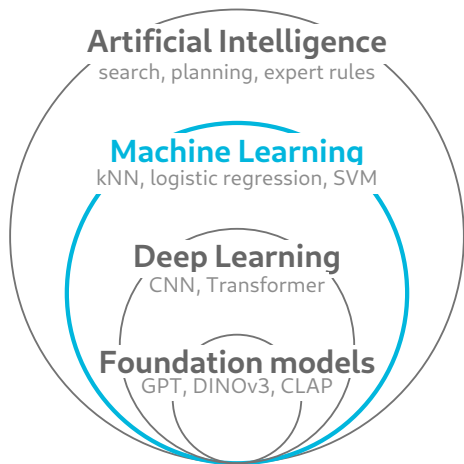
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Some key open challenges (core AI research)



This is the main paradigm shift you need to keep in mind for the whole course, and take home after. Until roughly 2020, researchers spent their energy fighting overfitting: "how do we make a model trained on a small dataset behave correctly on data it has never seen?" Generalization was THE problem. Foundation models flipped this: because they are pre-trained on internet-scale data, a generic, broadly capable model is now something you download rather than something you build. What is left to you is particularization: your task, your data, your language, your latency and memory budget. That is a different skill set, that we will cover in session 5. Note that this is an open research challenge, not a solved engineering recipe! It will move fast in the next years.

Let's dive into details!

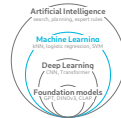


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Examples

- Learning to play chess through playing games,
- Learning to recognize dogs and cats in images from annotated examples ...

Machine learning

- **Supervised:** Learning from **inputs** and **human annotations**
- **Unsupervised:** Learning from **inputs** only (patterns)
- **Self-supervised:** Learning by reconstructing the **inputs** from distorted versions

Generalization

- Generalization refers to the ability to infer **good decisions or representations** from **examples**.
- Goal: transfer knowledge from a dataset to another

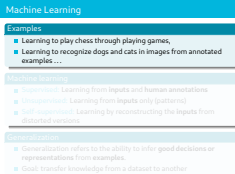
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└ Machine Learning

The most common framework in ML is Supervised Learning: we learn a function from inputs/examples and their corresponding outputs/labels. Here, labels are provided by humans (e.g. pictures of cats annotated as "cat"). A second framework is Unsupervised Learning, where the function is learned from the data alone, typically by finding repeated patterns or structure in it. A third, increasingly popular framework leverages the data itself to generate the annotations, removing the human from the loop; this can be very powerful when a relevant way of computing annotations is found. A typical example, used for most Large Language Models, is to remove a word from a sentence and train the network to predict it.

One key challenge in ML is Generalization: the ability to perform well on unseen data, different from the data (inputs/labels) the model was trained on. In this course, we will focus on the techniques used to build systems that learn from data and generalize to unseen data.



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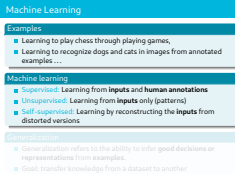
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Input/output

- **Goal:** infer a function of parameters $\mathbf{W}$ from an input (often tensor) space to an output (often tensor) space, $\hat{\mathbf{y}} = f(\mathbf{x}, \mathbf{W})$.
- **Example:** input can be an image, output a vector where the largest value indicate the category the image belongs to.

Error/Loss

- **Loss $\mathcal{L}$:** nonnegative measure of the discrepancy between expected output $\mathbf{y}$ and obtained output $\hat{\mathbf{y}}$.
- **Example:** $\mathbf{y} = [0, 1]$ but $\hat{\mathbf{y}} = [0.2, 0.8]$.

Parameters

- $f(\cdot, \mathbf{W})$ contains **parameters $\mathbf{W}$** to be trained,
- In most cases, an ideal $f(\cdot, \mathbf{W})$ exists but is **hard to find in practice**,
- Learning is a **regression ill-posed** problem.

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Global formalism

Loss: a way of telling the model when it is wrong, so it can be trained accordingly. The model contains parameters (weights and biases), and for a given task an optimal set of parameters usually exists; however, finding it is again an ill-posed problem, since many solutions may exist.

Global formalism

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- **Example:** $\mathbf{y} = [0, 1]$ but $\hat{\mathbf{y}} = [0.2, 0.8]$.

Parameters

- $f(\cdot, \mathbf{W})$ contains **parameters $\mathbf{W}$** to be trained,
- In most cases, an ideal $f(\cdot, \mathbf{W})$ exists but is **hard to find in practice**,
- Learning is a **regression ill-posed** problem.

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Course 1: Generalities about AI

└ Global formalism

Loss: a way of telling the model when it is wrong, so it can be trained accordingly. The model contains parameters (weights and biases), and for a given task an optimal set of parameters usually exists; however, finding it is again an ill-posed problem, since many solutions may exist.

Global formalism

Input/output

- **Goal:** infer a function of parameters $\mathbf{W}$ from an input (often tensor) space to an output (often tensor) space, $\hat{\mathbf{y}} = f(\mathbf{x}, \mathbf{W})$.
- **Example:** input can be an image, output a vector where the largest value indicate the category the image belongs to.

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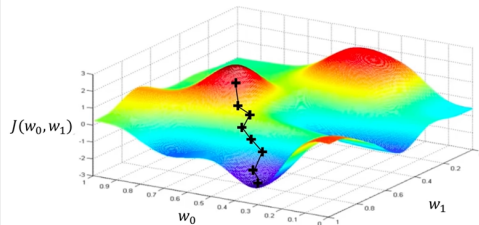
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- **Loss:** $J(\mathbf{W}) = \sum_i \mathcal{L}(f(\mathbf{x}^{(i)}, \mathbf{W}), \mathbf{y}^{(i)})$, $i = \text{examples}$
- Model parameters: $\mathbf{W}^* = \text{argmin}(J(\mathbf{W}))$

Training Algorithm

Gradient Descent:

- Randomly Initialize model weights
- Compute Gradient of the Loss $\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$
- Update weights
 $\mathbf{W} \leftarrow \mathbf{W} - \eta \frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$
- Repeat until convergence



from MIT course introtodeeplearning.com

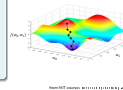
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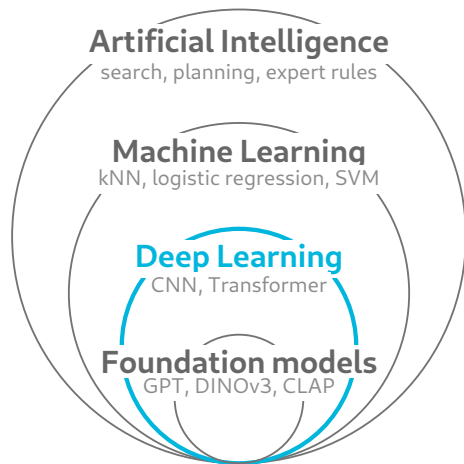
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The total loss J (also called Empirical Risk, or objective function) is the average loss over all inputs/examples, and the optimal model parameters are those that minimize it. But how do we find them? In other words, how do we train the model? Here is a simplified description of the training algorithm underlying modern DL: gradient descent, repeated until convergence to a local minimum (as illustrated for a simple example with only 2 parameters). We will see that the loss landscape becomes far more complex for models with millions of parameters, as in modern neural networks.

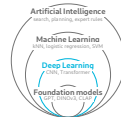


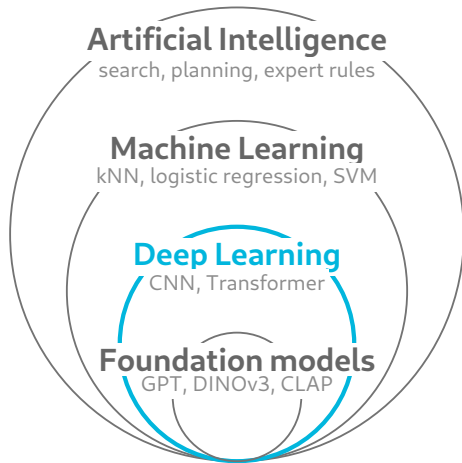
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Deep Learning

Deep Learning is best understood as a family of models (neural networks with many layers) rather than a learning framework in itself. It can therefore be trained in a supervised, unsupervised, or self-supervised way, exactly like any other Machine Learning approach.





A particular case of Machine Learning

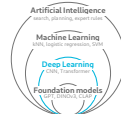
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└ Deep Learning

Deep Learning



A particular case of Machine Learning

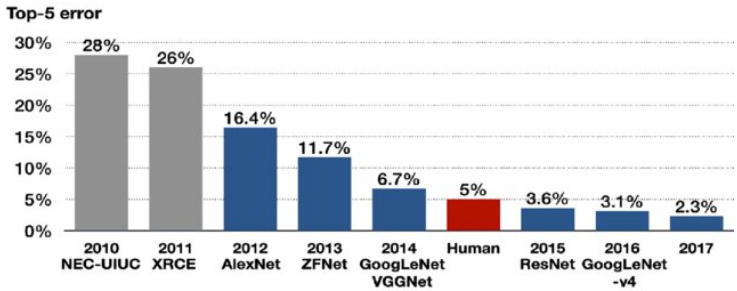
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A turning point: AlexNet (2012)

Started the reign of Deep Learning

- Using **deep** neural networks and **large** datasets
- A major breakthrough in image classification:

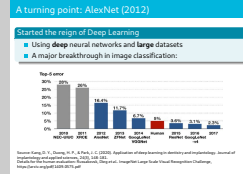


Source: Kang, D. Y., Duong, H. P., & Park, J. C. (2020). Application of deep learning in dentistry and implantology. Journal of implantology and applied sciences, 24(3), 148-181.
Details for the human evaluation: Russakovsky, Dieg et al.. ImageNet Large Scale Visual Recognition Challenge, <https://arxiv.org/pdf/1409.0575.pdf>

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└ A turning point: AlexNet (2012)



The landscape of Machine Learning changed in 2012: Deep Neural Networks, a technique used only in a minority of cases until then, suddenly won the image classification contest on ImageNet, a standard benchmark dataset. From that point on, Deep Neural Networks became mainstream, and their performance skyrocketed.

The great elders of Deep Learning (Turing Prize 2018)

Geoffrey Hinton



- Cognitive psychologist and computer scientist,
- Prof. Univ. Toronto / worked at Google until 2023,
- Known for back-propagation and Boltzmann machines (**Nobel Prize in 2024!**).

Yoshua Bengio



- Computer scientist,
- Prof. Univ. Montréal and founder of MILA,
- Known for his work on deep learning.

Yann le Cun



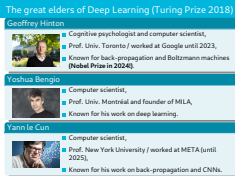
- Computer scientist,
- Prof. New York University / worked at META (until 2025),
- Known for his work on back-propagation and CNNs.

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└ The great elders of Deep Learning (Turing Prize 2018)

We mention them because their work largely helped the field emerge from the last two AI winters. It is also worth noting that they jointly received the Turing Award in 2018, the highest distinction in computer science.



Where did the Deep Learning revolution come from?

- The use of GPUs for computation.
- The share of huge datasets on Internet.
- HuggingFace / Github / Arxiv: easy and open-source share of research.
- The rise of representation learning.



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Where did the Deep Learning revolution come from?

DL training algorithms are highly parallelizable, and can therefore take full advantage of modern GPUs.

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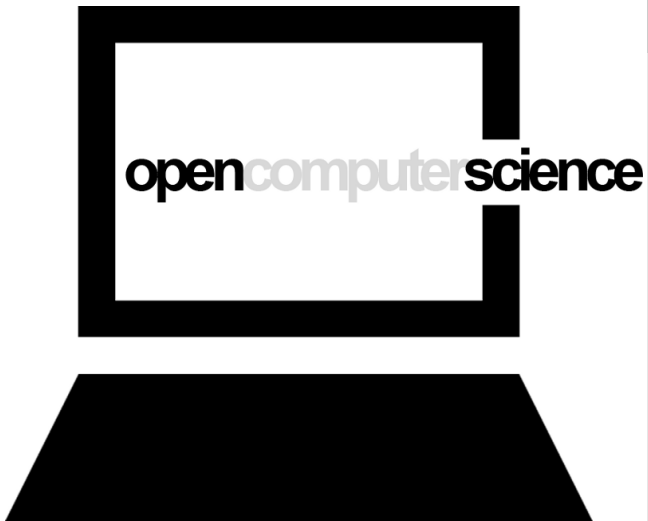
We live in a data-pervasive era: massive amounts of digital data are available and widely shared. This benefits DL algorithms, which are extremely data-hungry.

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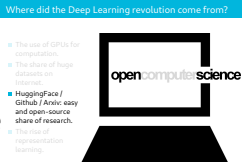


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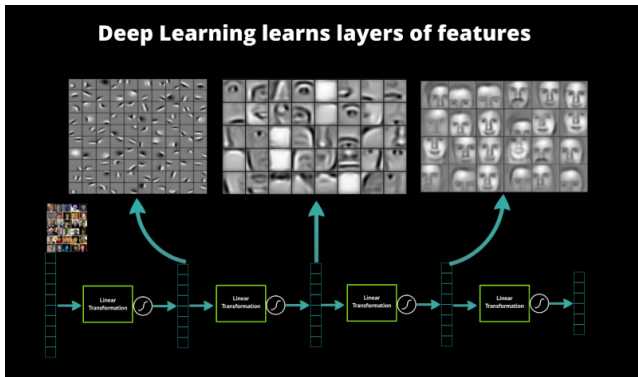
└ Where did the Deep Learning revolution come from?

The availability of open-source toolboxes such as PyTorch and Tensor-Flow, together with the practice of openly sharing research and code, makes implementing and training DL models much easier.



Where did the Deep Learning revolution come from?

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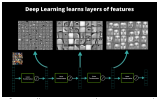
Source: <https://www.datasciencecentral.com/a-primer-on-deep-learning/>

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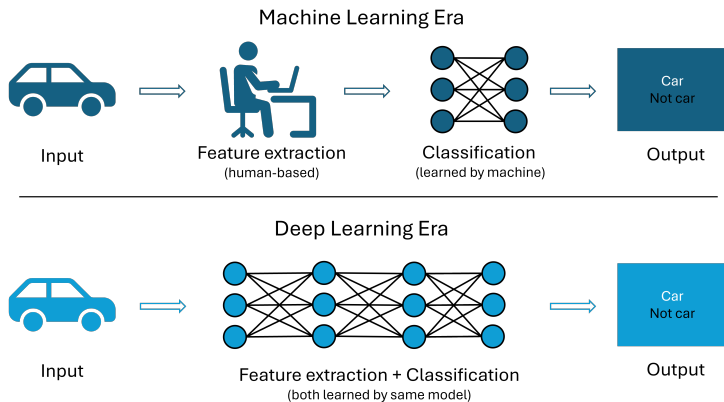
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A few more words on Representation Learning. As we will see in the next lessons, finding a good representation of the data is a genuinely hard problem. Deep Learning has made it possible to search for, and decompose data into, such representations automatically. In traditional pattern recognition, features were extracted using a priori expert knowledge about the data (e.g. recognizing oranges by filtering for the color orange and detecting round shapes). In mainstream pattern recognition, mathematical functions (e.g. wavelets) automatically decompose images into more abstract sets of features, though these are still chosen by experts depending on the data. In Deep Learning, by contrast, the features themselves are learned end-to-end.

Traditional Deep Learning (before 2020)

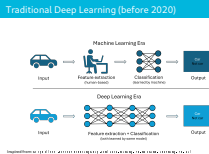


Inspired from: <https://www.softwaretestinghelp.com/data-mining-vs-machine-learning-vs-ai/>

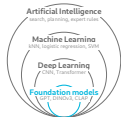
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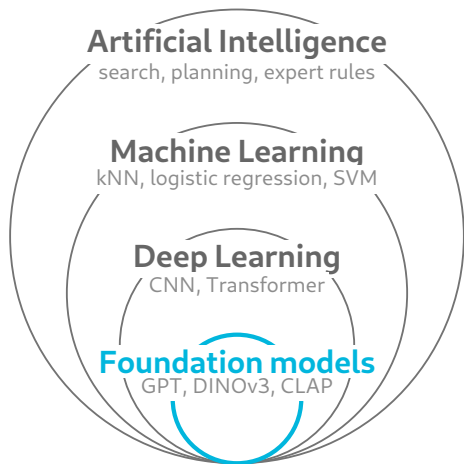
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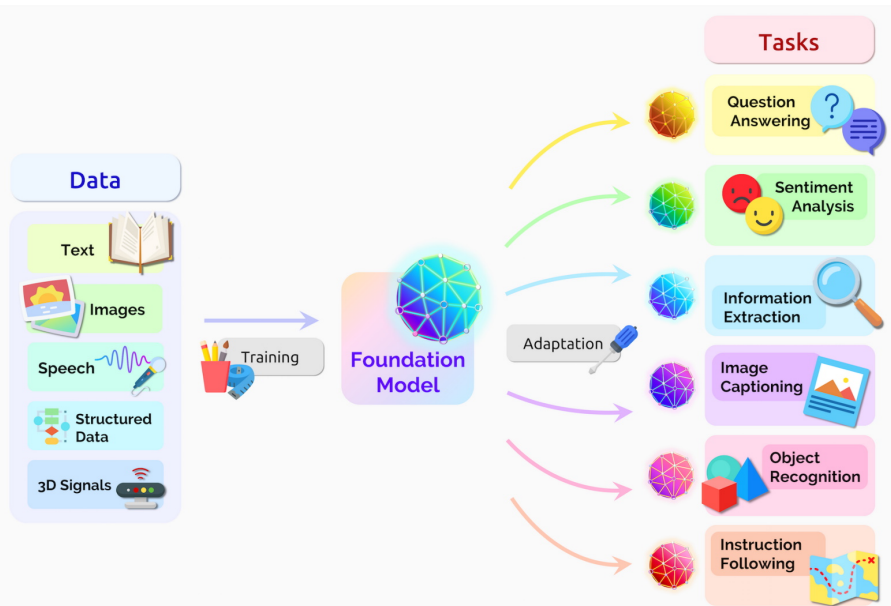
Before 2020, the dominant paradigm was to train one model per task, from scratch, on a dataset collected specifically for that task. This is the setting we will keep in mind for most of the technical content of this course, before contrasting it with the foundation-model paradigm.



└ Foundation models: a paradigm shift



Foundation models: a paradigm shift

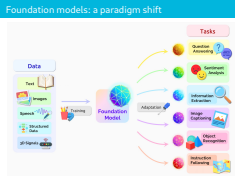


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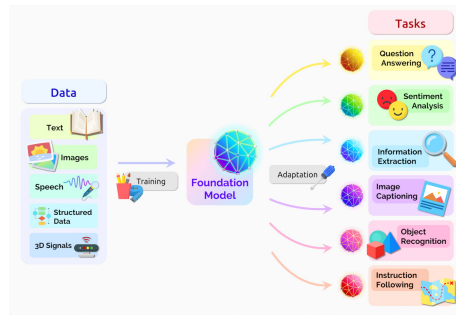
Foundation models: a paradigm shift

Foundation models are trained once, on internet-scale data, using self-supervised pretext tasks (i.e. tasks for which labels can be generated automatically from the data itself). Because of the scale of the pre-training data, generalization is essentially solved: the real challenge shifts to particularization, i.e. adapting the model to your specific problem.



Most recent Generative AI models are based on **Foundation models**:

- Model trained on an Internet scale dataset
- Self-Supervised training: pretext tasks
- Generalization is not a problem anymore! All is about **particularization**



Source: <https://blogs.nvidia.com/blog/what-are-foundation-models/>

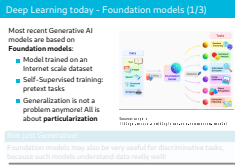
Not just Generative!

Foundation models may also be very useful for discriminative tasks, because such models understand data really well!

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└ Deep Learning today - Foundation models (1/3)

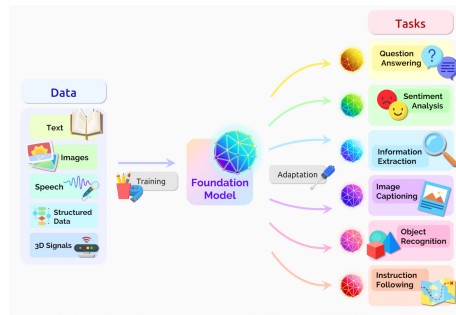


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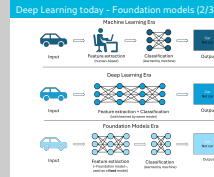
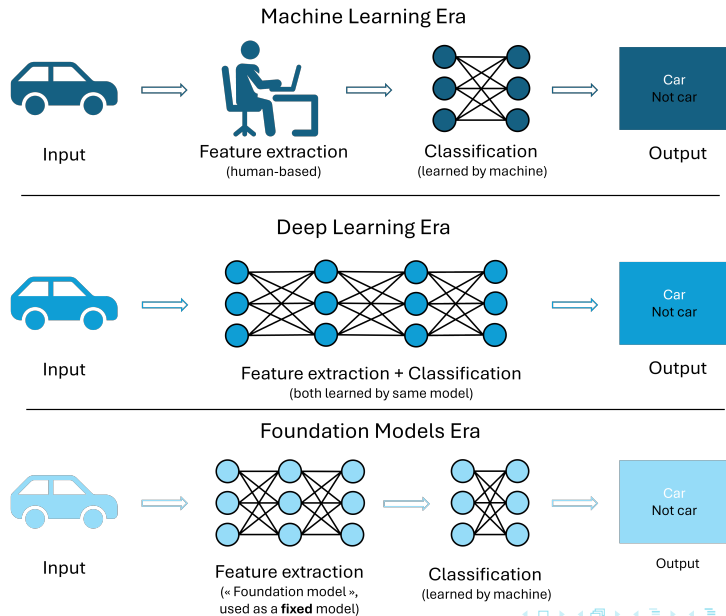
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Deep Learning today - Foundation models (2/3)

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Deep Learning today - Foundation models (2/3)



This figure contrasts the traditional pipeline (one model trained per task) with the foundation-model pipeline (one large pre-trained model, then lightly adapted to many tasks). Keep this picture in mind, as it summarizes the paradigm shift emphasized in this course.

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Deep Learning today - Foundation models (3/3)

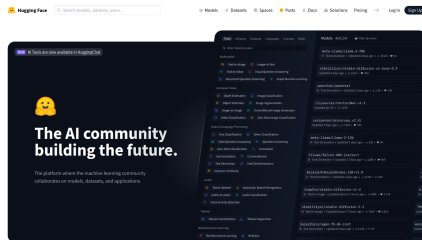
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 - Requires a lot of time
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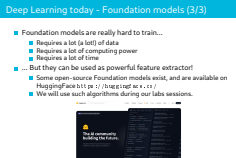


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Lab Session 1 (1/2) — Essential tools for data science

What you will learn from the lab:

- A dataset is a set of **points in a space** ;
- **Distances** between those points already reveal structure ;
- Sometimes, data comes broken, and you need to **clean it** first.

Lab assignments — 1h15:

- Python, IDE, virtual environments,
- Introduction to **Numpy**: arrays, shapes, broadcasting,
- Generate two clouds of points, plot them, compute the **pairwise distance matrix**, interpret it,
- Load a **real** dataset and describe it: size, missing values, annotated or not, class balance, ...

Link to the lab:

<https://mee-labs.gitlab-pages.imt-atlantique.fr/intro2ai/lab1/>

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Course 1: Generalities about AI

└ Lab Session 1 (1/2) — Essential tools for data science

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Lab Session 1 (2/2) — Your project (this afternoon)

The project is the **backbone** of this course

- **2h30 this afternoon:** form your binôme, find a dataset, define one or two tasks ;
- You will present your project **on next session**. This will be evaluated!



Your dataset must be...

- $\approx 1\,000-10\,000$ samples,
- as **complete** as possible,
- in an **easy-to-use** format,
- under an **open license**.

Post your project title on Discord before you leave!

Topics are unique within a group, reserved first-come.

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Course 1: Generalities about AI

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Timeline: S1 (define today) → S2 (validate graded) → S3 → End (final presentation)

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